

Candidate Profile Transformation Pipeline — Technical Design

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Goal. Ingest heterogeneous candidate records from multiple sources, resolve which belong to the same person, merge into one canonical profile per person with per-field provenance + confidence, and emit through a runtime-configurable projection. Three invariants everywhere: **never fabricate a value**; **never crash the batch** on one bad record (skip + log); **deterministic & explainable** output.

1. Pipeline

inputs → **adapters** → **normalize** → **resolve identity** → **merge (+confidence)** → **project (+validate)** → **JSON**, with a **RunReport** (skips / conflicts / assumptions / counts) threaded through every stage.

- **Adapt** — one pluggable SourceAdapter per source (recruiter CSV, ATS JSON, GitHub REST, résumé PDF) → a normalized SourceRecord (raw + normalized per field). load() is try/except-wrapped: a bad source/record becomes a skip, not a crash.
- **Normalize (deterministic)** — phone→E.164, dates→YYYY|YYYY-MM, country→ISO-3166 α-2, skills→alias map, email→lowercase + validate.
- **Resolve identity** — O(n) blocking, then precision linking.
- **Merge + confidence** — one cluster → one CanonicalProfile of TrackedValues.
- **Project + validate** — the only config-aware stage; canonical → flat, validated dict.

2. Canonical schema & normalized formats

CanonicalProfile: **candidate_id** (deterministic SHA1 of the strongest stable anchor — email → phone → name-key, so reruns are identical), full_name, emails[], phones[] (confidence-sorted), location{city, region, country}, links{linkedin, github, portfolio, other[]}, headline, skills[] {name, confidence, sources[], experience[], education[], repos[], years_experience, overall_confidence, flags[]}. Every value is a **TrackedValue**{value, confidence, sources[], provenance[], competitors[]}.

Formats. **Dates** YYYY or YYYY-MM — granularity preserved, “Present”/””→None (ongoing), never fabricate a month. **Phones** E.164 via phonenumbers — missing country code + --default-region → applied & flagged; no region → keep raw, no guess. **Country** ISO-3166 alpha-2 via pycountry; unresolved → null (raw kept). **Skills** alias-mapped, preserving + and # (C++/C#.NET never collapse to “c”); unknown kept verbatim at lower confidence + flag. **Email** lowercased + shape-validated; a bad value is dropped, not the record.

3. Merge / conflict-resolution policy

- **Match keys (union):** normalized email, github_login, name_block_key (sorted set of first letters of name tokens — survives initials/reordering, e.g. “Sri Krishna V” ≡ “V, Sri K.”). Exact email or login → link outright; else name-alignment + shared phone ≥ threshold. No pairwise fuzzy over all records (keeps it O(n)).
- **Single-valued winner** = highest source trust: **ATS 0.90 > CSV 0.80 > résumé 0.75 > GitHub 0.70**. Losers retained as competitors (never discarded) + Flag(conflict_resolved) + a ConflictEntry.
- **Multi-valued** (skills, emails, phones) = union + dedup, each value scored independently.
- **Confidence** = clamp01(base + corroboration – extraction – conflict) × recency: base = winner trust; corroboration +0.05 per extra agreeing source × independence (ATS↔CSV 0.5, any↔GitHub 1.0), cap +0.10; extraction –0.10 for prose/heuristic; conflict –0.05; recency decays only time-varying fields via last_updated. **Overall** = weighted Σ over core fields (name .25, email .20, phone .15, company .15, title .15, location .10); absent field → 0.

4. Runtime custom-output config (projection + validation)

A JSON config is an ordered list of **FieldSpec**: **path** (output key), **from** (canonical path — field, field.sub, field[].sub; defaults to path), **type** (string | string[] | number | object | object[]) — drives a dynamic model), **required**, **normalize (assert-only:** verify the value already matches E164/iso3166-a2/canonical; a mismatch is treated as missing; the projector never recomputes — avoiding double-normalization/drift), **on_missing** (null | omit | error), and include_confidence/include_provenance; plus global on_missing + include_flags. The **Projector is the only config-aware component** (hard boundary at CanonicalProfile). Output is validated by a Pydantic model built **at runtime** from the config (create_model); a profile that fails validation is skipped + logged, batch continues. The default config reproduces the required schema verbatim; a custom config renames/reshapes/toggles confidence + provenance **with zero code change**.

5. Edge cases & deliberate descopes

Handled. (a) Company conflict (ATS *Stripe* vs CSV *Shopify*) → trust picks the winner, losers kept as competitors, conflict_penalty + flag. **(b) Name variants, no shared email** (*Sri Krishna V* / *Vijayarajan*) → blocking on name_block_key + github_login collapses them to one cluster; a GitHub record with no email/login and a non-aligning name stays an **orphan** (documented). **(c) Garbage source** — malformed CSV/JSON, corrupt PDF, or a scanned (no-text) résumé → adapter try/except → RunReport skip, other sources still emit. **(d) Phone without country code** → --default-region + assume:default_region + Flag(assumed_region); with no region the raw is kept and never guessed. **(e) Skill alias + unknown** → ReactJS→React, C++/C#.NET preserved, unknown verbatim @0.60 + flag.

Deliberately deferred under time pressure (all pluggable behind existing seams): live LinkedIn / recruiter-notes adapters; résumé experience/education parsing (lean scope today); fuzzy/embedding **name & skill** matching (deterministic alias + blocking chosen for explainability); embedding skill-similarity for the long tail (design noted in the README).